Worksheet: Logic – Part 2 (Topic 1, Part 2)

Introduction

Welcome back. In this module we move from building truth tables to *using* them: we learn that two formulae can be **equivalent** even when they look different, and we collect a small toolbox of equivalences (De Morgan's laws, the implication equivalence and the contrapositive) that let us rewrite and simplify logical statements without drawing a single table. We then take a big step into **first-order logic**, where predicates describe properties of objects and quantifiers let us reason about *all* or *some* of them at once.

By the end of this worksheet you will be able to:

- recognise and prove propositional equivalences, including De Morgan's laws, $(p \to q) \equiv (\neg p \vee q)$ and the contrapositive;
- rewrite a formula using only conjunction and negation;
- translate everyday English into quantified first-order statements and judge them true or false with a witness or a counterexample;
- correctly negate quantified statements.

A motivating taste of where this leads: on an island of knights (always truthful) and liars (always false), person A says "at least one of us is a liar" and person B says nothing. By the end you will have the tools to work out who is who.

Estimated time: approximately 60 minutes.

> **How to use this worksheet.** Work the activities in order, from basic to more challenging. Make a genuine attempt at each one on paper before you look at the answer sheet at the end; the answers are fully worked out so you can check your reasoning line by line.

Key terms and notation

Term / symbol	Meaning
$\equiv$	**Equivalence**: $A \equiv B$ means A and B have identical truth tables (the same truth value for every assignment). It is a *relation* between formulae, **not** a connective you can use inside a formula.
$\neg$	Negation ("not").
$\wedge$, $\vee$, $\to$	Conjunction ("and"), disjunction ("or"), implication ("if … then … ").
De Morgan's laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$. The negation is pushed inside, every proposition is negated, and the connective flips.

Term / symbol	Meaning
Implication equivalence	$(p \to q) \equiv (\neg p \lor q)$, and therefore also $\equiv \neg(p \land \neg q)$.
Contrapositive	The contrapositive of $(p \to q)$ is $(\neg q \to \neg p)$; the two are equivalent.
Tautology / contradiction / consistent	A **tautology** is true for every assignment; a **contradiction** is false for every assignment; a formula is **consistent** if it is not a contradiction (i.e., it is true for at least one assignment).
Predicate	A property of one or more objects, e.g. Odd(3) or Equal$(5, 6)$. Once given its arguments it becomes a proposition (true or false).
Domain D	The set of objects the quantifiers range over, e.g. $D = \{x_1, x_2, \ldots, x_n\}$.
$\exists$	**Existential quantifier**: $\exists x\, P(x)$ means "for some x, $P(x)$ is true". One witness is enough.
$\forall$	**Universal quantifier**: $\forall x\, P(x)$ means "for all x, $P(x)$ is true". It must hold for every element.
Witness / counterexample	A single value that makes an $\exists$-statement true, or a single value that makes a $\forall$-statement false.

> **Common pitfall.** "All P's are Q's" translates as $\forall x\,(P(x) \to Q(x))$, **not** $\forall x\,(P(x) \land Q(x))$. The conjunction would claim that *every* object is both P and Q. Likewise "some P's are Q's" uses $\land$, not $\to$.

Activities

Activity 1

Start point. De Morgan's laws let you push a negation through a bracket: $\neg(p \land q) \equiv \neg p \lor \neg q$ and $\neg(p \lor q) \equiv \neg p \land \neg q$. You negate every proposition *and* flip the connective. As a model, here is the truth table for the first law (using 1 for true, 0 for false):

p	q	$\neg(p \land q)$	$\neg p$	$\neg q$	$\neg p \lor \neg q$
1	1	0	0	0	0
1	0	1	0	1	1
0	1	1	1	0	1

p	q	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
0	0	1	1	1	1

The two outer columns match, which is what the equivalence claims.

Instructions.

1. Negate each statement using De Morgan's law, pushing the negation all the way in (the first is a simple example).
   - (a) "It is Wednesday and it is not sunny."
   - (b) "It is raining or the ground is dry."
   - (c) "It is not cold and it is not windy."
2. Build the complete truth table for the **second** law, $\neg(p \vee q) \equiv \neg p \wedge \neg q$, with columns p, q, $\neg(p \vee q)$, $\neg p$, $\neg q$, $\neg p \wedge \neg q$. Confirm the two outer columns are identical. State whether the column for $\neg(p \vee q)$ is a tautology, a contradiction, or neither.

   Hint. For (c) remember that negating "not cold" simply gives "cold"; double negation cancels.

Expected output. Three negated English sentences, and one fully filled four-row truth table with a classification of the $\neg(p \vee q)$ column.

Answers: see Activity 1 in the answer sheet.

Activity 2

Start point. Two further equivalences let you change the shape of an implication. First, $(p \to q) \equiv (\neg p \vee q)$; applying De Morgan's law to the right-hand side also gives $(p \to q) \equiv \neg(p \wedge \neg q)$. Second, the **contrapositive**: $(p \to q) \equiv (\neg q \to \neg p)$.

Instructions.

1. Write $(p \to q)$ using **only** conjunction and negation.
2. State the contrapositive of $(p \to q)$.
3. Show, as a chain of equivalences, why $(p \to q) \equiv (\neg q \to \neg p)$, using $(p \to q) \equiv (\neg p \vee q)$ and the fact that $\vee$ is commutative.
4. Give the contrapositive, in English, of: "If it rains, then the ground is wet."

Expected output. One formula for part 1, one formula for part 2, a three- or-four-step equivalence chain for part 3, and one English sentence for part 4.

Answers: see Activity 2 in the answer sheet.

Activity 3

Start point. Because $\vee$ and $\to$ can each be expressed with $\wedge$ and $\neg$ (namely $p \vee q \equiv \neg(\neg p \wedge \neg q)$ and $p \to q \equiv \neg(p \wedge \neg q)$), *any* formula can be rewritten using

only conjunction and negation.

Instructions.

1. Rewrite $p \vee (q \to r)$ using only $\wedge$ and $\neg$, showing each step and naming the equivalence you use.
2. (Recall) State how a conjunction $p \wedge q$ can be written using only $\vee$ and $\neg$.

   Hint. Work from the outside in: convert the outermost $\vee$ first, then the $\to$, then any remaining $\vee$.

Expected output. A multi-step rewrite ending in a formula containing only $\wedge$ and $\neg$, plus one short formula for part 2.

Answers: see Activity 3 in the answer sheet.

Activity 4

Start point. In first-order logic a **predicate** such as Prime(x) describes a property of an object x chosen from a **domain** D. Quantifiers let us talk about the whole domain: $\exists x\, P(x)$ ("some x satisfies P") needs just one witness, while $\forall x\, P(x)$ ("every x satisfies P") must hold for all of them. The figure below shows a tiny domain and how the two quantifiers behave on it.

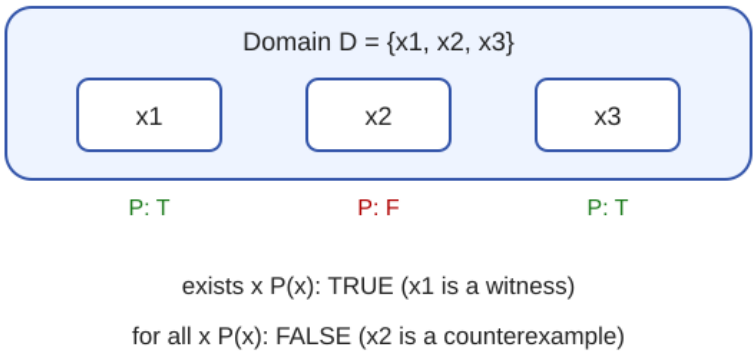


Figure 1

Figure 1. On a three-element domain, one true case is enough for the existential quantifier, but a single false case defeats the universal quantifier.

Notice that the existential statement is true because at least one element works, whereas the universal statement is false because one element fails.

Instructions.

1. First, using the figure, state the truth values of $\exists x\, P(x)$ and $\forall x\, P(x)$, and identify the witness and counterexample.
2. Take the domain to be the integers, with predicates Prime(x), Odd(x), Even(x), Mult4(x) ("x is a multiple of 4") and Mult2(x) ("x is a multiple of 2"). For each English sentence, (i) write the quantified formula and (ii) state whether it is true or false, giving a witness (for true $\exists$-statements) or a counterexample (for false $\forall$-statements).
   - (a) "All prime numbers are odd."
   - (b) "No prime numbers are even."
   - (c) "Some prime numbers are even."
   - (d) "Some prime numbers are not even."
   - (e) "All multiples of 4 are multiples of 2."

Expected output. One line for the figure, and five rows, each with one quantified formula and a true/false verdict plus a witness or counterexample (or a one-line justification for a true $\forall$-statement).

Answers: see Activity 4 in the answer sheet.

Activity 5

Start point. Quantifiers can be negated by an extension of De Morgan's law: $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$ and $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$. The quantifier flips and the negation moves inside.

Instructions.

1. Negate $\forall x\, (P(x) \to Q(x))$, showing each step, and give the final simplified formula.
2. Without a truth table, prove the equivalence $\neg\exists x\, (P(x) \wedge Q(x)) \equiv \forall x\, (P(x) \to \neg Q(x))$ as a chain of equivalences.

   Hint. In part 1, after flipping $\forall$ to $\exists$, replace the implication with $\neg P(x) \vee Q(x)$ before pushing the negation in with De Morgan's law.

Expected output. A four-step negation ending in one formula, and a four-step equivalence chain.

Answers: see Activity 5 in the answer sheet.

Wrap-up

You have now used truth tables to *verify* equivalences, applied De Morgan's laws, the implication equivalence and the contrapositive to *rewrite* formulae, converted whole expressions to a single pair of operators, and moved into first-order logic to translate English, judge quantified statements with witnesses and counterexamples, and negate quantifiers. These are the core tools of formal proof: turning a claim into a precise formula and then manipulating that formula with guaranteed-equivalent steps. They connect directly to what comes next in

the course, where you will use exactly this kind of equivalence-driven reasoning to build and check proofs.

As a capstone (optional, outside the time budget), return to the island riddle from the introduction.

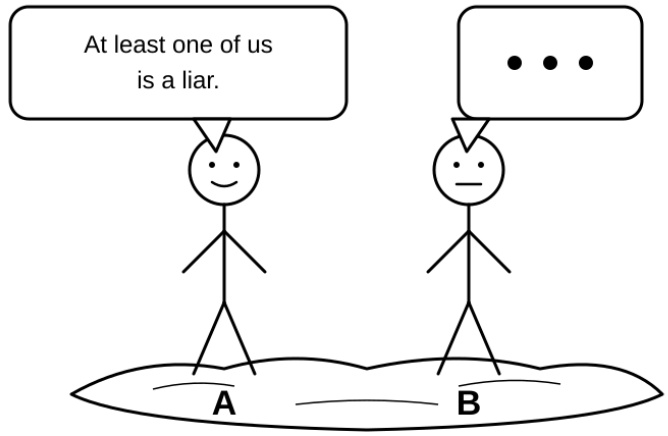


Figure 2

Figure 2. Person A makes a claim about the pair; person B is silent. The logic of the claim alone determines who is a knight and who is a liar.

Reason it out: if A were a liar, then "at least one of us is a liar" would be false, meaning *no one* is a liar — a contradiction, since A would be one. So A must be a knight, making the statement true; because A is a knight, the "at least one liar" must be B. Hence **A is a knight and B is a liar**, even though B said nothing.

Answer sheet

Activity 1 (answer)

1. Negations:
   - (a) "It is **not** Wednesday **or** it is sunny." (Negate "Wednesday", flip $\wedge$ to $\vee$, negate "not sunny" to "sunny".)
   - (b) "It is **not** raining **and** the ground is **not** dry."
   - (c) "It is cold **or** it is windy." (Each double negation cancels; $\wedge$ flips to $\vee$.)
2. Truth table for $\neg(p \vee q) \equiv \neg p \wedge \neg q$:

p	q	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
1	1	0	0	0	0
1	0	0	0	1	0
0	1	0	1	0	0
0	0	1	1	1	1

The columns $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are both $0, 0, 0, 1$, so the equivalence holds. The column for $\neg(p \vee q)$ is **neither** a tautology (not all 1s) nor a contradiction (not all 0s).

> **Gotcha.** A frequent error is to write $\neg(p \wedge q) \equiv \neg p \wedge \neg q$, forgetting to flip the connective. The negation must change $\wedge$ into $\vee$ (and vice versa); otherwise the two sides disagree on the rows where exactly one of p, q is true.

Activity 2 (answer)

1. $(p \to q) \equiv \neg(p \wedge \neg q)$.
2. Contrapositive: $(\neg q \to \neg p)$.
3. Chain:
$$(p \to q) \equiv (\neg p \vee q) \equiv (q \vee \neg p) \equiv (\neg q \to \neg p).$$
(The middle step uses commutativity of $\vee$; the last step applies $(a \to b) \equiv (\neg a \vee b)$ with $a = \neg q$, $b = \neg p$.)
4. "If the ground is not wet, then it did not rain."

Activity 3 (answer)

1. $p \vee (q \to r)$:
$$\begin{aligned} p \vee (q \to r) &\equiv \neg(\neg p \wedge \neg(q \to r)) \\ &\equiv \neg(\neg p \wedge \neg(\neg q \vee r)) \\ &\equiv \neg(\neg p \wedge (q \wedge \neg r)). \end{aligned}$$
Final: $\neg(\neg p \wedge (q \wedge \neg r))$.
2. $p \wedge q \equiv \neg(\neg p \vee \neg q)$.

> **Gotcha.** When converting a $\vee$ that sits *inside* a negation, do not drop the negation. In part 1 the term $\neg(\neg q \vee r)$ becomes $q \wedge \neg r$ (De Morgan), not $\neg q \vee r$; losing the outer negation changes the whole formula.

Activity 4 (answer)

1. From the figure: $\exists x\, P(x)$ is **True** (witness x_1); $\forall x\, P(x)$ is **False** (counterexample x_2).

2.
- (a) $\forall x\,(\text{Prime}(x) \to \text{Odd}(x))$. **False**; counterexample $x = 2$ (prime but not odd).
- (b) $\forall x\,(\text{Prime}(x) \to \neg\text{Even}(x))$. **False**; counterexample $x = 2$ (prime and even).
- (c) $\exists x\,(\text{Prime}(x) \wedge \text{Even}(x))$. **True**; witness $x = 2$.
- (d) $\exists x\,(\text{Prime}(x) \wedge \neg\text{Even}(x))$. **True**; witness $x = 3$.
- (e) $\forall x\,(\text{Mult4}(x) \to \text{Mult2}(x))$. **True**; every multiple of 4 is divisible by 2, so no counterexample exists.

Gotcha. "All P's are Q's" must use $\to$, not $\wedge$. Writing $\forall x\,(\text{Prime}(x) \wedge \text{Odd}(x))$ would assert that *every* integer is both prime and odd, which is plainly false for a different reason and is not what the English says.

Activity 5 (answer)

1. Negating $\forall x\,(P(x) \to Q(x))$:

$$\begin{aligned}\neg\forall x\,(P(x) \to Q(x)) &\equiv \exists x\,\neg(P(x) \to Q(x)) \\ &\equiv \exists x\,\neg(\neg P(x) \vee Q(x)) \\ &\equiv \exists x\,(P(x) \wedge \neg Q(x)).\end{aligned}$$

   Final: $\exists x\,(P(x) \wedge \neg Q(x))$.
2. Proving $\neg\exists x\,(P(x) \wedge Q(x)) \equiv \forall x\,(P(x) \to \neg Q(x))$:

$$\begin{aligned}\neg\exists x\,(P(x) \wedge Q(x)) &\equiv \forall x\,\neg(P(x) \wedge Q(x)) \\ &\equiv \forall x\,(\neg P(x) \vee \neg Q(x)) \\ &\equiv \forall x\,(P(x) \to \neg Q(x)).\end{aligned}$$

Gotcha. In part 1 the most common slip is to negate the implication as $\neg P(x) \wedge Q(x)$. The correct push-in of $\neg(\neg P(x) \vee Q(x))$ via De Morgan's law gives $P(x) \wedge \neg Q(x)$: the antecedent keeps its sign and the *consequent* is the part that gets negated.